

HNDIT23103

**Advance Database
Management Systems**

**Week 03
Relational Model**

Learning Outcomes

- At the end of the lesson you should be able to describe the fundamental concept of Relational Data Model and its operations

THE RELATIONAL MODEL

- The relational model is a broad, flexible model
- Basis for almost all DBMS products
- E.F. Codd defined well-structured “normal forms” of relations, “normalization”

RELATION 2016

- A is a Two-dimensional table containing Rows (*tuples*) and Columns (*attributes*) of data.

Student	<u>Reg_No</u>	Name	ContactNo	Address	Gender	Age	GPA
	1111	C.D. Perera	0372228222	Kurunegala	M	20	3.9
	1112	C. Basnayake	0362255222	Awissawella	M	21	3.5
	1114	N.S. Bandara	0812234567	Kandy	F	20	3.8
	1115	K. Peris	0112876655	Colombo	M	21	3.0
	1116	S. Menike	0253456654	Anuradhapu ra	F	21	2.6

Relational Model -Properties

- Each relation (or table) in a database has a **unique name**
- An entry at the intersection of **each row and column is atomic** (or single-valued); there can be no multi-valued attributes in a relation
- **Each row is unique**; no two rows in a relation are identical
- Each attribute (or column) within a table has a unique name
- The sequence of columns (left to right) is insignificant; the columns of a relation can be interchanged without changing the meaning or use of the relation
- The sequence of rows (top to bottom) is insignificant; rows of a relation may be interchanged or stored in any sequence

Relational Model 2013

- *The relational model of data has three major components:*
- **Relational Database Objects**- allows to define data structures
- **Relational Operators** – allows manipulation of stored data
- **Relational Integrity Constraints** – allows to define business rules and ensure data integrity

Equivalent Relational Terms

Informal Terms	Formal Terms
Table	Relation
Column	Attribute/Domain
Row	Tuple
Values in a column	Domain
Table Definition	Schema of a Relation
Populated Table	Extension

ATTRIBUTE 2013

- Each column in the relation is an attribute of the relation.
- The number of attributes of the relation is called the **DEGREE OF THE RELATION**.
- The set of possible values that an attribute may have is the **DOMAIN** of the attribute.

Entity 2013

- An **entity** is an object in the real world

TUPLE

- A tuple is a collection of values that makes up one row of relation.
- A tuple is the equivalent of a record.
- Number of tuples in a relation is its **CARDINALITY**.

Relational Model

Name of the Relation

Primary Key

Attributes

Student	<u>Reg_No</u>	Name	ContactNo	Address	Gender	Age	GPA
	1111	C.D. Perera	0372228222	Kurunegala	M	20	3.9
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Tupels

Domain

Degree of the Relation - 7

Cardinality Of the Relation - 5

KEYS

- A key of a relation is an attribute in a relation.
- There are various types of keys in a relation.

PRIMARY KEY

- The attribute (or attributes in combination) for which no more than one tuple may have the same value is called the primary key.
- That is a value that can be used to uniquely identify each record.
- The relational data model requires that the primary key of a tuple should not contain null values.
- An attribute selected from the candidate keys to be used to uniquely identify each row

CANDIDATE KEY

- Candidate keys are all attributes that may serve as the primary key.
(one key is selected as a primary key)

ALTERNATE KEY

- The candidate keys other than the chosen **primary key** are called **alternate keys**.

COMPOSITE/CONCATENATED KEY

2013

- A key that consists of two or more attributes appended (joined) together.
- A primary key tat is made up of two or more attributes

RELATIONAL INTEGRITY CONSTRAINTS

- Constraints are *conditions* that must hold on *all* valid relation instances. **There are three main types of constraints:**
 - **Key** constraints
 - **Entity integrity** constraints
 - **Referential integrity** constraints

Integrity Rules

Entity integrity

- Requirement (Ensures all entities are unique): all primary key entries are unique; no null value
- Each entity has unique key

Referential integrity

- Foreign key must match primary key values
- Makes it impossible to delete row whose primary key has mandatory matching foreign key values in another table

KEY CONSTRAINTS

- **Superkey** of R: A set of attributes SK of R such that no two tuples *in any valid relation instance $r(R)$* will have the same value for SK. That is, for any distinct tuples t_1 and t_2 in $r(R)$,
 - $t_1[SK] \neq t_2[SK]$.
- **Key** of R: A "minimal" superkey; that is, a superkey K such that removal of any attribute from K results in a set of attributes that is not a superkey.

KEY CONSTRAINTS

- **Example**: The VEHICAL relation schema:
- VEHICAL(State, Reg#, SerialNo, Make, Model, Year)
- has two keys Key1 = {State, Reg#}, Key2 = {SerialNo}, which are also superkeys. {SerialNo, Make} is a superkey but *not* a key.
- If a relation has *several* candidate **keys**, one is chosen arbitrarily to be the **primary key**. The primary key attributes are *underlined*.

ENTITY INTEGRITY

- **Relational Database Schema:** A set S of relation schemas that belong to the same database. S is the *name* of the **database**.
- $S = \{R_1, R_2, \dots, R_n\}$
- **Entity Integrity:** The *primary key attributes* PK of each relation schema R in S cannot have null values in any tuple of $r(R)$. This is because primary key values are used to *identify* the individual tuples.
- $t[PK] \neq \text{null}$ for any tuple t in $r(R)$

REFERENTIAL INTEGRITY

- A constraint involving *two relations* (the previous constraints involve a *single* relation).
- Used to specify a *relationship* among tuples in two relations: the **referencing relation** and the **referenced relation**.

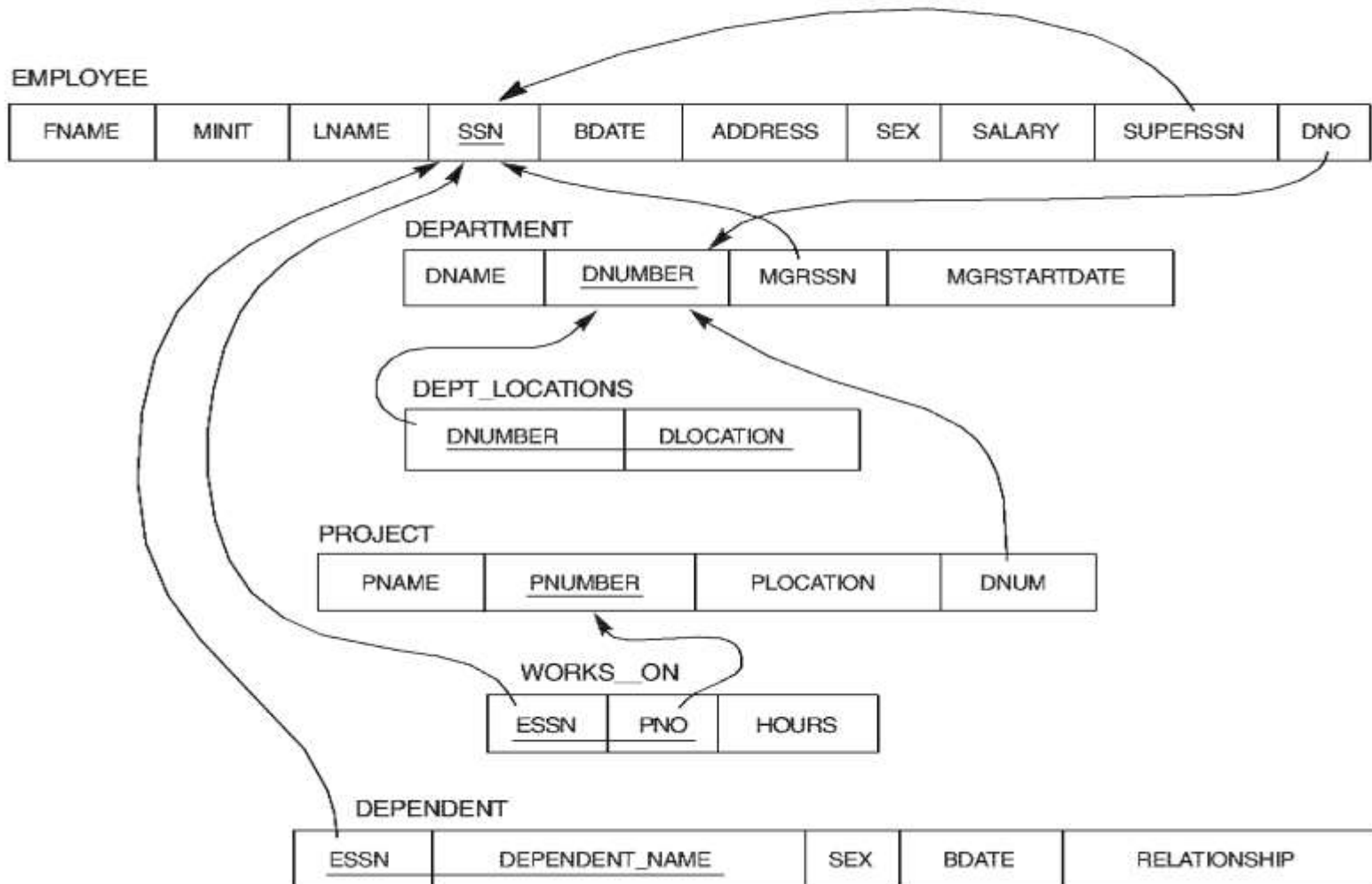
REFERENTIAL INTEGRITY

- Tuples in the *referencing relation* R_1 have attributes FK (called **foreign key** attributes) that reference the primary key attributes PK of the *referenced relation* R_2 . A tuple t_1 in R_1 is said to **reference** a tuple t_2 in R_2 if $t_1[\text{FK}] = t_2[\text{PK}]$.
- A referential integrity constraint can be displayed in a relational database schema as a directed arc from $R_1.\text{FK}$ to R_2 .

REFERENTIAL INTEGRITY CONSTRAINT

- Statement of the constraint
- The value in the foreign key column (or columns) FK of the **referencing relation** R_1 can be either:
 - (1) a value of an existing primary key value of the corresponding primary key PK in the **referenced relation** R_2 , or..
 - (2) a null.
- In case (2), the FK in R_1 should not be a part of its own primary key.

Example - Referential integrity constraints displayed on the COMPANY relational database schema diagram.



IDENTIFIER ATTRIBUTE OR KEY ATTRIBUTE

<i>Key</i>	• Data item that allows us to uniquely identify individual occurrences or an entity type.
<i>Superkey</i>	• Attribute or set of attributes that uniquely identify a tuple.
<i>Candidate key</i>	• Minimal super key with the property of irredusability and uniqueness
<i>Primary key</i>	• An entity type may have one or more possible candidate keys, the one which is selected as primary key.
<i>Composite key</i>	• candidate key that consisting of two or more attributes
<i>Foreign key</i>	• An attribute or set of attribute that matches the candidate key or other or same relation

The name of each primary key attribute is underlined.